

ActInf Textbook Group ~ Cohort 3 ~ Meeting 6

(Chapter 3, part 1)

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hello and welcome everyone it's March 1st 2023 we're in cohort three and we are having our first discussion today on chapter three chapter three um is the high road to active inference previously in chapter two we talked about the low road and today we're talking about the high road those two roads are laid out in figure 1.2 we took it from Bayes theorem to active inference in chapter two and chapter three is going to be starting from the free energy principle and also taking us to octave inference so before we go through the chapter or look at any questions does anyone want to just give any reflection or thought what was some experience they had in reading chapter 3 or a quote or an idea or something that stood out for them you can raise your hand or just go for it okay so feel free to raise your hand or write in the chat if you ever want to add anything um chapter three takes a different attack than chapter two [Music] anyone want anything or we'll we'll start exploring through the chapter and then coming to the questions that people have raised um today or in the next week okay we'll move through chapter three hopefully relatively quickly to get to the first of the questions and then we'll have seeded some questions for coming to our discussion next week so in section 3.1 in chapter 2 free energy was motivated as a bound in a way that we can do approximate Bayesian inference coming from the low road to active inference and you can do free energy based or energy-based methods like a variational autoencoder you could do that on any Bayesian statistical problem at all and it's used commonly it doesn't necessarily mean that it's connected to action selection at all in chapter 3 they're going to start from The High Road which is this Central imperative that organisms or really just things must maintain their existence which is going to be operationalized in a way that's compatible with the surprise minimization that was brought up in chapter two and just like in Chapter 2 where free energy was proposed as a way to bound surprise which is like what we would really want to know coming from the high road we also want to bound surprise and it turns out that minimization of free energy is again going to appear as a computational computationally tractable solution to the problem the chapter is

going to describe the formal equivalence between the minimization of variational free energy and maximization of model evidence or self-evidencing in Bayesian inference and then active inference is going to be brought up because we're going to head into chapter 4 which is going to be all about the generative models at the heart of active inference active inference is about how organisms or adaptive systems maintain their existence by minimizing surprise via more proximally this tractable proxy variational free energy and the two ways to minimize variational free energy are perception and action change your mind change the world integrated imperative that supports this unifying perspective on systems high road is going to start from the premise that any living organism or any persistently observed thing has to maintain itself in a set of preferred States if it doesn't maintain itself in a set of expected slash preferred States it just isn't that kind of thing that is going to apply to everything from an oil droplet that's diffusing or not to more sophisticated cognitive goals humans physiological entities have to stay within their physiological ranges in reward learning it might be proposed that there's this value function where acceptable ranges are more rewarding and then rewarding Paths of action would be selected in contrast in active inference we establish the expectations around those physiological variables in terms of a preference and then we reduce our surprise about realizing those preferences slash expectations that allows us to take the most likely course of action to maintain being the kind of thing we are rather than proposing this reward function and then asking what's the most rewarding outcome foreign active inference cast the biological problem of or explanation for survival as surprise minimization surely that will spur many conversations but for now this is not like surprise birthday minimization so this isn't to say that surprise doesn't happen or that cognitive surprise is not even sought after actively in certain cases this is referring to the statistical concept of surprise which is defined for a very specific generative model so not just surprise in general but for a given generative model at whatever level it's constructed at surprise minimization is what leads to the maximization of model evidence because if you can minimize surprise or the tractable proxy of variational free energy your model will basically be as fit as it can be against any thoughts or just raise your hand and we can take it anyway 3.2 mark off blankets does anyone want to just give a preemptive thought on Marco blankets or anything else Jonathan go for it sorry it's actually with respect to the surprise and and this is sort of with respect to some preference distribution over the the status of the um of the agent presumably does the agent have to have any awareness of that preference distribution itself and can the preference distribution itself be something which is inferred um you know we may

infer something about our preferred States um you know uh is there a sort of hierarchy to that as well nice questions does anyone want to give a first thought does the agent need awareness of preference and can preferences be learned okay just first passes uh Ali go for uh yeah actually I believe uh it can be modeled um in a kind of uh nested hierarchical way so that uh the agent can be aware of the aware of its own preferences and also preferences uh is not uh static at all and they can be dynamic as well and can be learned and updated great exactly if the model does have a higher nested level that enables the interpretation of awareness on preferences sure in the base model no no awareness attention phenomenology qualia is ever implied at the base level so it doesn't have to be but you could make a model that has that feature can preferences be learned again in the most simple case no the C Vector is fixed however there have been several works by Sajid at all and others on preference learning will oh I feel like you said most of what I was just about to say just that you know thinking at the level of a bacterium you know awareness is a a tricky concept but um clearly there is an embedding of these preferences in in you know some information at whatever level we're talking about and yeah talking about learning preferences you know perhaps at the highest level those don't really change but if you're talking about lower level preferences um those are kind of adapted to to the individual uh time span like the bacteria may not um kind of change its Behavior overall on on the on a lifespan but uh you know given an individual couple of minutes it's going to say oh I prefer to go left versus right because of the chemical gradient great yes so a preference slash expectation for being in the Run versus the tumble mode of movements is happening faster and might be learnable or updatable whereas a slower updating process of for example whether sugar is preferred at all might be happening at a different time scale um ever and Francois yes I'm also thinking about Michael Levin's work you had him on one of your um episode former episodes where he's talking about basal cognition and that there's maybe even something like nowhere of preferences at the very very lowest level cool thank you Francois and then Jonathan and the example that that will said that you know lower level preferences could change like moving left or right for the bacterium how do we know if that's a preference or not just like it's re-evaluating different policies that would lead to different expected free Energies yeah it's a great question if we don't Define the system like it's not a system that we created then it would be an empirical question whether the data support a model in which you have a fixed preference with oscillating policy selection based upon expected free energy calculations or whether there are behavior that seems to go beyond that to actually capture uh a plastic C so it's an empirical question from a given data set and from a

given because we can easily imagine and just speak out both of those examples and so it's like a portfolio of models and then we would do model selection or evaluation amongst those alternative strategies um Jonathan and then Evert yeah I mean is there some um some way of thinking about it that there's always going to be a hierarchy that there is simply a preference for survival within within any model like that that um you know the agent is trying to retain its thickness um and from that maybe glucose levels and temperature and you know what whatever else it may be but each of those is a preference in sort of uh in order to have this macro preference which is simply survival that's a great way to put it over intra and especially intergenerational time scales you're only going to ever see things that at least act that way or appear that way that's kind of like the good regulator theorem from cybernetics so yeah at some point it can be said that there's this implicit pervasive preference for persistence otherwise you just wouldn't be even observing that system to be anyway at all ever yeah would you say there's a difference between preferences and goal directed Behavior it's a great question in fact goal is not used in the active inference ontology we can definitely talk informally about like well the preference is to be on Square three comma 3 in the grid world so just Loosely speaking its goal is to make it to three comma three but we don't need to appeal to goals as a formal construct nor reward so conversationally it's totally fair to talk about uh agentic or goal directed Behavior the way that that's being operationalized in active inference is through this preference vector thanks Scott um always might be off there are presumably bad Regulators which maintain themselves by luck and bad Regulators which will lose their homeostasis imminently yes totally agreed with that luck is only going to go so far and so long and the fact that the imperative is for adaptive regulation doesn't mean there aren't systems that fail this is probably also related to what we discussed before and as if talk that the agent looks as if it's minimizing this or that or it looks as if it's pursuing a certain goal would you say that's right yes as if or perhaps a synonym for as if could be modeled as or I could publish a paper modeling it as as a very um uh defined way of talking about as if I can imagine it or it could be written that or one could model that and that puts us more squarely in the instrumentalist perspective or just the imaginative perspective so awesome comments great from surprised with respect oh um I'll ego for it uh yeah I just wanted to point out uh perhaps we can talk about gold-directed motion as a kind of uh emergent uh phenomena that arises from this energy minimization mechanism so it's not by itself an objective parameter that we need to describe or model it's just uh it can be seen or described as an emergent phenomenon of that mechanism awesome and I just uh Everett go for it sorry I just had to like re-share my

screen I'm sorry I had sorry for my handset all right so continuing on with Section 3.2 big topic in the field the way that things are going to be modeled or framed is in terms of their relative insulation from the environment the generative process the environment and the relative insulation with the generative model or the agent the particular States in the absence of the separation there would be no surprise to minimize there must be something to be surprised and something to be surprised about Unity is plural at minimum two in other words there are at least two things system and environment and these can be disambiguated from one another these can be modeled as being disambiguated from one another a formal way to express that separation is the statistical construct of a Markov blanket which can be traced to the work of Judea Pearl on Bayesian graphs and causal here is box 3.1 with a technical definition of a Markov blanket a Markov blanket in brief they write is a set of variables that mediate all statistical interactions between a system and its environment so Markov blankets are nodes one or more nodes on a Bayesian graph for which two other sets of nodes are conditionally independent so if person A says something to person B and B says something to C it's like person B is the blanket between A and C so it isn't the internal external States can't influence each other but rather it's that their engagements with each other are mediated through the blankets and um just earlier uh it's in chapter nine we had a really great discussion about the map territory fallacy and the map territory fallacy fallacy and for now just to sort of see this discussion and continue um the movement Markov blankets exist for generative models because Markov blankets are a feature of Bayesian graphs so the world the territory does not have Markov blankets however Maps can have Markov blankets and a huge amount of well intended and off-key discussion has lost touch with that map territory distinction Markov blankets are about Maps because we're talking about partition States on a base graph the claim is not that the real world consists of labeled things that have internal external sense and action on them and then just one more point and again feel free to raise your hand or anything in the discussions of pearl the Markov blanket is this insulating set of nodes what Friston and collaborators have done in the previous two decades or so is further partition that undirected Markov blanket into two types of nodes nodes with incoming dependencies which are interpreted as sensory States and nodes with outgoing dependencies which are active States that licensed the use of the Markov blankets to describe the statistical boundaries of an Adaptive or cybernetic agents so the Markov blanket concept itself is undirected it just describes the insulating set of nodes Friston at all operationalized it further to bring it into the setting of cognitive modeling so now we revisit that particular partition that previously was

shown as a feedback loop with perception and cognition and action and now it's going to be shown in a way that's going to be taking us closer to the Bayesian mechanics or the particular partition which is we have the blanket which is shown here according to the the Friston blanket approach active states with outgoing statistical dependencies sensory states with incoming statistical dependencies internal states don't have a direct edge to external states so there's no telekinesis and there's no telepathy however internal states can select actions that modify how the external world changes that influences which sensory states are observed which is interpreted through perceptions learning by internal states so consequences percolate throughout the entire partition but there's no direct connections between external and internal states Ali uh yeah I just wanted to quickly point out um about two issues here one of which is um I think an important error in this figure uh because uh well we know that the flow of active states uh wouldn't depend on the external states and similarly the flow of the internal sensory states wouldn't will not depend on the internal state so in the equations in that figure X and Y should be interchanged yeah that's it and the other one is of course to emphasize my favorite definition of Markov blanket which is the uh it's a kind of boundary or interface in state space so it's not necessarily a spatial temporal boundary although uh in some cases it might manifest itself as a spatial temporal boundary such as the membranes of cells or organelles or whatever great way to say it the Markov blanket is the boundary in the statistical state space okay so um I'm just gonna add some of these questions uh Jonathan I'm just adding it to the chapter three questions and we'll we'll um come to it I'm just having this weird thing with like freezing of my screen um and then ever yeah I just posted the link in the chat I don't know if it's possible to open the page yes but because there isn't a picture a little bit down I think that's also a mark of blanket that's shown there but there are more um more arrows than in the one that you just showed in the textbook a little bit lower yeah so if you and I wonder yeah so I wonder no it's not that one okay which page oh I have a look uh it's on page let's figure one okay I'm gonna have to reshare again sorry okay I hope that this is a little bit more stable the uh well just a few possible comments any base graph is going to have a multitude of Markov blankets Markov blankets are not uniquely identifying like agent environment boundaries but those are some boundaries in state space that we're interested in but every base graph that isn't fully connected so if there's any nodes that aren't connected to other nodes they will have a Markov blanket so the concept is actually less philosophically and even semantically loaded than people often expect any base graph you just draw on a piece of paper if there's some node that intermediates two other sets of nodes it's a Markov blanket it does not

identify cybernetic system boundaries with respect to the cybernetic system boundary though which is one kind of base graph that we care about the work of Aguilera at all how particular is the free energy principle has explored different topologies of the particular partition for example like what if you do or don't have this Edge between active and sensory States why is there an arrow going from action back to internal why is there an arrow going from sense back to external the first answer on this is that this is a statistical model so just because you're enabling that edge doesn't mean that the data are going to support that edge having a non-zero parameterization so just describing the possibilities that are going to get parameterized so it's not a claim about the real world #map territory it's just saying this is the class of models that were fitting so in practice it means less than it may seem however it's also the case that what constraints we put on this topology do end up influencing greatly what kinds of models we fit Michael yeah I have a question on page 43 under this box it says illustrates the interpretation of marker blanket in a dynamic setting and then the sentence here the conditional independencies have been supplemented with dynamical constraints so that the flow do not depend on states on the opposite opposite side of the blanket so can you elaborate what that means and what this implies uh What uh dynamical constraints in contrast to what else the the dynamical constraints are the dynamical equations that are provided for the rates of change of each of these and again note what um what Ali has brought up in and written in the chat um there there's just a slight error with the the variables as written so where it says that $\dot{U}$ so that's like action States it says that it's a function of X U and Y um it should be μU and Y because this would suggest that octave states do depend on external States when actually this x should be a μ which is to say that active States are a function of internal States sensory states and their own current position that doesn't mean that they're all equal contributors it's empirical with your data set and your generative model and you can imagine situations where it's only the internal state that matters or it's only the sensory state that matters this is just like the space in which models are fit from this isn't to Claim about any specific cybernetic system or any interpretation of the real world octopus uh how so x -star indicates that you're dealing with continuous time Dynamics correct these are flows on the variables Okay so um I'm having a tough time mapping this is there any disjunction or any difficulty with things that are inherently discrete uh like I love cellular automata and any discrete agent-based model can be put in a cellular automaton framework but there's no way to really I mean you can I guess you can interpolate it somehow but it's not a natural way of thinking about it so is continuous time essential to this framework or is it just a convenient much of the textbook

thank you octopus is about the differences and similarities between discrete time and continuous time formalizations so just to give a quick preview figure 4.3 shows on the top The partially observable Markov decision process discrete time discrete State space formalization and the bottom is the continuous time formalization which is more like a Taylor series approximation using the generalized coordinates of motion and then chapter 7 focuses on the discrete time generative models chapter 8 focuses on the continuous time models and in figure 8.6 there's a nested hierarchical model where the lower level is continuous time and the higher level is discrete time generative models can be either discrete or continuous State spaces time or other variables uh Jonathan wrote presumably in reality all these variables are high dimensional vectors for sure it could be as simple as a scalar a single number or it could be an n-dimensional tensor ever yeah thanks octopus efforts are Henry's or or anyone else no I didn't have my hands okay it was just my zoom was locking okay is going to give an intuition for what it means when we have this um a simple system with μ internal x external and B blanket States so we can imagine that we're sampling from this Cube X Y and Z but it's μ B and uh and X and then these three panels are X conditioned on B μ conditioned on B and then finally the relationship between X and μ and so this shows that you could estimate X conditioned upon an expectation of μ given B so internal States given blanket States that's like cognition about X so here's my estimate my posterior probability estimate of X which I'm not directly observing conditioned upon my expectations of internal States conditioned on B and so you can have a relationship between X and B like there's some correlation between external and blanket States some correlation between the temperature in the room and your thermometer and then there's some correlation between your cognitive States and the if your beliefs about temperature given the thermometer and then that results in there being an information relationship between external and internal States though they are only mediated through the blanket measurements this is just an inference passive setting there's no action coming into play yet it's just to get a bit of an intuition about how a Markov blanket mediated perceptual inference occurs 3.3 surprise minimization and self-evidency so an agent with Markov blanket appears to model the external environment in the sense that internal States correspond on average to a probabilistic representation of external states of the system figure 3.2 that's what that was showing importantly the agent's generative model cannot simply mimic external Dynamics otherwise the agent would simply follow external dissipative Dynamics so passive might lead one to want a model where we simply estimate external States as well as we can we just learn optimally and just sit and watch the show rather the

model must also specify the preferred conditions for the agent's existence or the regions of states that the agent has to visit to maintain its existence or satisfy criteria for its existence in terms of occupying characteristic States so if we want to do more than just watch if we want to act to bound our surprise and improve our persistence in a dissipative far from equilibrium environments we need to have some preferences or expectations that actions can be selected based upon how they will realize those preferences Francois yeah sorry I think we kind of skipped over something but I didn't really understand we want to ask about um is that the thing about the the Symmetry across the market plan Mark of blanket and the consequences of this being weakened um associate pairs of expected internal and external states with each other I really don't understand how that consequence comes from the Symmetry question mark of blanket could and could you explain it better in a different way sure there's more um angles let's just say first starting with a homeostatic angle let's just say that temperature really is relevant for so the color of the room doesn't matter for survival but the temperature matters and we have a thermometer so that's our Markov blanket and we're going to be getting some proxied information about temperature through our thermometer without going too deep into debates about representationalism there must be something in the generative model the agent the cognitive model of the agents that represents that temperature there has to be some mapping between the temperature in the room unobserved and the agent's perception of temperature such that it's able to engage adaptively otherwise it's just going to be a particle adrift so it turns out it doesn't need to have an exact structural relationship so temperature could be a continuous variable and the agent's generative model might just be too hot just right or cold three states so I don't need to have the same structure but there does have to be a mapping between decisions that influence vital and the external reality of those vital variables so that's the homeostatic angle and then just shortly the other symmetry based angle is um the external states don't have to be an active agent but they could be this could be like a Communications interface in which case it would be fair to say that the generative model performs inference with an aboutness of the generative process but also the process as agent is doing inference on the aboutness of the agent on the right side so that has been more recently explored with like a symmetry with a niche doing inference on the organism also through the same blanket okay thank you that makes more sense great so that um baking in of or expectations builds in what they call an implicit optimism bias which is like steering our perception and action towards realizing preferences about observations this optimism bias is necessary for the agent to go beyond the mere duplication of external Dynamics or

resemblance of external Dynamics to prescribe active states that underwrite its preferred or characteristic States one can cast optimal Behavior with respect to Prior preferences which again doesn't mean that it always works doesn't mean everything always lives all the time or that outcomes are the best that they can ever possibly be but given prior preferences the maximization of model evidence by perception is a signal processing perspective that's like the optimal reconstruction of some sound however we have a unified imperative that isn't just maximizing model evidence for perception we have a joint imperative for the maximization of model evidence by perception and action we can reduce our surprise bound our with minimizing variational free energy minimizing expected free energy which is a dual or unified imperative that brings together changing your mind through perception learning and changing the world through action um a good fit indicates the model successfully accounts for Sensations at the same time it realizes its preferred sensation given that they are less surprising good description good prescription good scientist good engineer such a good fit is a guarantee of surprise minimization as maximizing model evidence $P(Y)$ the probability of the data given the model is mathematically equivalent to minimizing surprise fancy I negative natural log of the same that is why you'll often hear adaptive systems described as self-evidencing self-evidencing means acting to Garner sensory data consistent with an internal model hence maximizing model evidence surprise minimization as a hamiltonian principle of least action this is starting to invoke some more physics based approaches on the low road we were squarely within statistics Bayes theorem base theorem with action active inference coming from the free energy principle on the high road we have a lot more physics based views expressing this as a physicist might dot dot this is like a ball rolling down a hill from a high gravitational potential energy at the top of the hill to low energy in a basin so just like in classical mechanics it might describe a ball rolling down an elevation landscape with respect to potential energy we're in an information geometric landscape and the ball rolling down the hill taking the path of least action is going to be formally equivalent or at least formally analogous to our belief distributions rolling down the hill with negative log evidence or surprise playing the role of gravity that is what is equivalent to potential energy drawing the ball down the hill and if you want a more technical angle check out live stream 52 which will be happening this week and next week it is going to go a lot deeper into this surprise as potential energy interpretation here's a path taken by a two-dimensional random dynamical system it's kind of vibrating but it is also spending most of its time in the sensor and the left one another way to view like the landscape that this pen was drawing on or that this ball was

rolling on is on the right this is like a topo map where here is like a plateau and then it's like a bowl so even though it started here it kind of vibrated around but it spends most of its time in the bottom of the bowl in the middle is a trajectory of a system that has Dynamics bearing no relationship to surprise so here you have a uncorrelated just random walk Brownian diffusion inner particle more details and equations that connect minimization of entropy which is kind of like the um blurriness of a distribution like how broad or sharp it is has a relationship with the maximization of the probability of observations and also the expectation of self-information so ensuring that a small proportion of sensory states are occupied with high probability like we want a distribution on temperature States that's very sharp that is equivalent to bounding surprise about temperature yes and also maintaining a certain entropy on the temperature distribution Maria does anyone know if the random pattern in the middle is less or more common than the centered one Jonathan wrote it depends on the context totally agreed you can add more details than that Maria but there are um sets of variables in in your imagination or in the world for whom their Dynamics look like that and there's sets that look like that and various other things in between surprise minimization formalizes the idea of homeostasis uh Ali go for it uh yeah I think it might be useful to point out uh Paul Curry's the taxonomy of singular points because uh back in the 19th century he showed that uh singular points can have only four possible different types uh namely nodes saddle points Foci and centers but then he went out uh claiming that uh among those different types nodes saddle points and Foci are much more generic than Center so centers only arises in exceptional circumstances so yeah in almost all the physical systems that we model physical dynamical systems uh we will most likely encounter only those three kinds of singular points um and it's just very briefly uh nodes are the points through which an infinite number of solution curves pass saddle points are the points through which only two solution curves pass and Foci are the points which the solution curves approach in the manner of the logarithmic spiral and the centers are the points around which the solution curves are somehow Inc closed and enveloping one another great just really quickly to efforts uh question so this sensory active state notice that the notation is totally different so you got to take a fresh look at the notation every time like external states are Q Prime and so on um so m is a blanket M and Φ are blanket between q and E but also e and Ω are a blanket between q and M so mark all blankets are everywhere they do no philosophical work they're a purely technical description of nodes on a Bayesian graph maps have Markov blankets not territories it can be somewhat misleading when we have a picture of a brain as if we were describing it but it's like actually a

map of the brain not the territory itself so brains don't have Markov blankets Maps do maps are made by people maps are made amongst portfolios of other maps and some of those Markov boundaries in Phase space can be identified with different functional phenomena but just drawing a base graph is not a prediction or an explanation by itself and so it doesn't do too much work yet a huge amount of discourse has focused on those types of topics ever yes do you do you agree that that's a big description of the Markov blanket that captures more uh maybe things that have been discussed in previous uh times about preferences and subjective feelings because I know that Mark songs has talked about precision as being related to the things that we consciously receive as feelings is that captured here in this picture do you see preferences are not shown here unless one uses preference as a subtype of internal State marks Holmes indeed has discussed extensively Precision affect Consciousness awareness and so on in the end it comes down to just the full specification of the generative model what data and what phenomena are being tried to be explained by that model and data together and just drawing any kind of um diagram is just the beginning of a graphical abstract or a schema it's not an explanation prediction design imperative or control system it's just a drawing all right in our last five minutes topics really interesting so this is our first discussion on and we got about halfway through we're just going to peep ahead and then next week we will um pick up with the chapter three questions already there's a bunch of them we've been talking about entropy self-suprimal model evidence maximization and their relationships 3.4 relations between or among depending on how you prefer your English inference cognition and stochastic Dynamics table 3.1 is going to juxtapose statistical physics Bayesian inference and information Theory and a cognitive interpretation where in rows there are going to be sort of three phases so across columns these three Fields as the authors are showing or talking about something that's basically like the same or resonant with box 3.2 is talking about free energy in statistical physics and active inference they want to be clear that variational free energy which is a used commonly in machine learning is not necessarily referring to for example the Gibbs free energy the actual liberatable energy of the particles constituting that embodied entity 3.4.1 variational free energy model evidence and surprise here surprise the information on the data given the is the negative log of model evidence and model evidence is always less than the variational free energy which we'll go into more 3.42 expected free energy and inference of most likely trajectories so variational free energy is real time it's about the model right now and the data right now expected free energy is going to be about likely courses of actions action sequences are called policies and that's going to entail the description of

observations that haven't actually happened yet Jack oh yeah I just wanted to ask a question regarding the distinction between Gibbs free energy and variational free energy as you just mentioned in box 3.2 so um is this just a metaphor between the two like an analogy or is there a mathematical like relationship that we can write down between Gibbs free energy and variational free energy yes there are formal relationships among structurally that that license the description of all of those as energy authors like Alex Kiefer have um argued Beyond where many go to talk about the um uh actual uh free energy minimizing features of Gibbs maintaining systems but let's return to that and then 3.5 active inference and novel Foundation to understand behavior and plain text section 3.6 models policies and trajectories plain text reconciliation of inactive cybernetic and predictive theories under active active inference emergence of life to agency plain text 3.9 summary living organisms visit characteristic States active inference thank you everybody we will return next week or in the office hours and just write your questions in the relevant pages and we'll continue the discussion so thanks a lot everybody for joining farewell